1. Safety

Always engage the emergency stop and wait for the spindle to coast to a stop (about 15 seconds) before opening the dust shoe.

Wear hearing protection: the spindle reaches 86 dB at 18,000 rpm.

2. Specifications

Working area 1220 x 1220 mm (48 x 48 in); Z travel 150 mm.

Spindle: 2.2 kW water-cooled, 6,000-24,000 rpm, ER20 collet.

Rapid traverse 15 m/min; positioning repeatability 0.03 mm.

3. Maintenance schedule

Every 8 hours: vacuum the gantry rails and wipe the linear guides with a lint-free cloth.

Every 40 hours: grease the linear bearings with NLGI 2 lithium grease, 2 pumps per bearing block.

Every 500 hours: replace the spindle coolant (50/50 distilled water and propylene glycol, 6 litres) and check the ball screw backlash; above 0.05 mm, replace the ball nut.

4. Alarm codes

ALM-3 - Spindle coolant flow low: the pump delivers less than 2 litres per minute. Check the reservoir level and the inlet strainer.

ALM-9 - Limit switch hit on X: jog the axis back off the switch in the opposite direction, then re-home.

ALM-14 - Spindle overheat above 45 C: stop the job, let the spindle cool for 20 minutes and check coolant flow.